

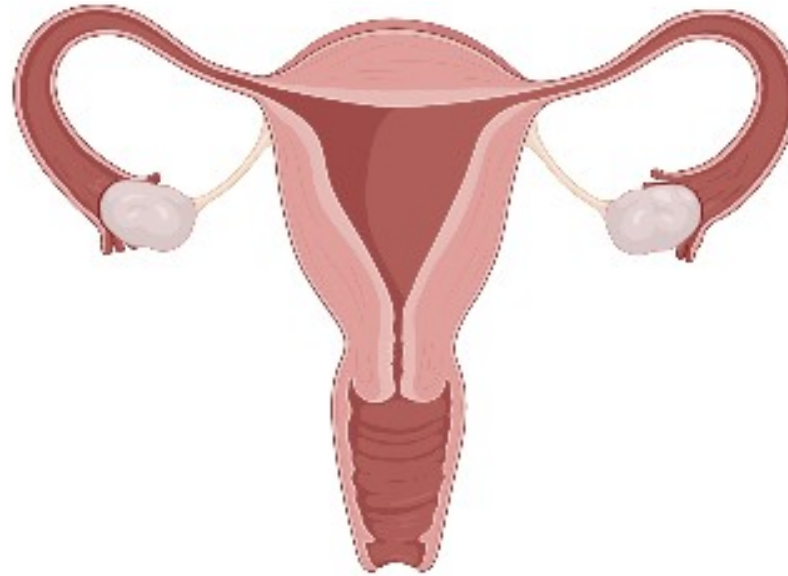
Pathophysiology

Screening

Management

Cases

Cervical Cancer Screening



Pathophysiology

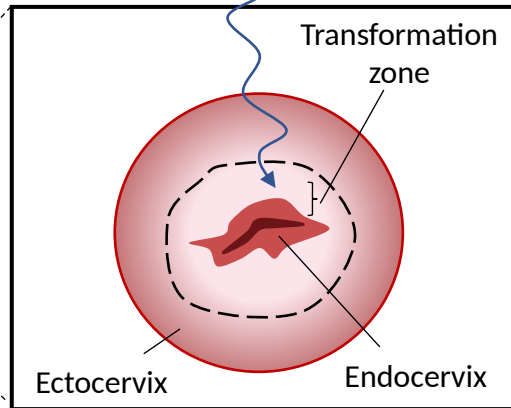
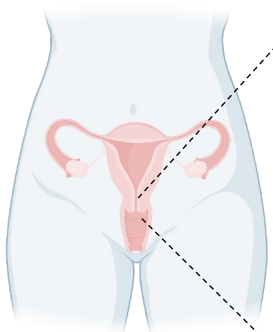
Screening

Management

Cases

What causes cervical cancer?

HPV
(e.g., subtypes 16, 18)



10%

Persistent HPV infection

~10 y

Cervical cancer

90%

Spontaneous clearance
(at 2 years)

Pathophysiology

Risk factors for persistent HPV infection?

- High-risk subtypes (16 and 18)
- Older age
- Immunosuppression
- Smoking
- Infection with other STIs
- cOCP use

Cervical anatomy

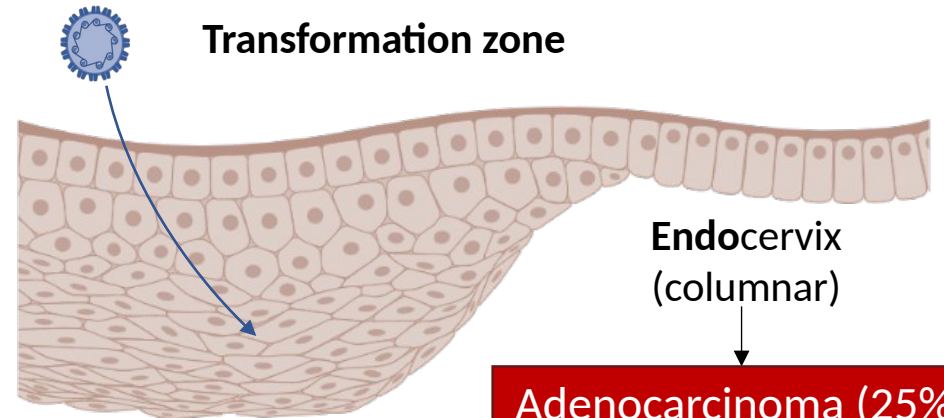
90% of cancers develop at the transformation zone

Cervical cancer types

Ectocervix
(squamous)

SCC (75%)

Transformation zone



Endocervix
(columnar)

Adenocarcinoma (25%)

Pathophysiology

Screening

Average risk

High risk

Management

Cases

Screening

Detects treatable changes and precancers that are likely to progress to cancer

Persistent
HPV infection

Atypia

Precancer

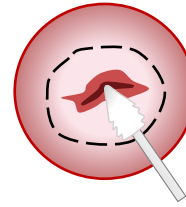
Cervical
cancer

Screening
tests?

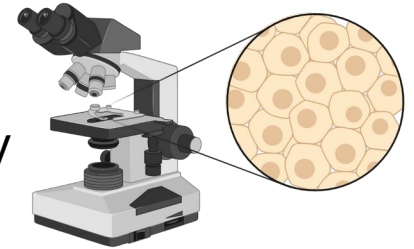
Cytology

+/-

Primary
HPV



A **Pap smear** collects exfoliated cervical cells, which are examined for changes suggestive of acute HPV infection, precancer, or cancer



Detects high risk types of HPV from a cervical sample

Pathophysiology

Screening

Average risk

High risk

Management

Cases

Average Risk Patients

Patients without a history of immunocompromise or CIN2/CIN3

USPSTF
2018

Age

20

30

40

50

60

Cytology
q3y

Cytology q3y | Co-test or primary HPV q5y

ACS 2020

Primary HPV q5y > Co-test q5y | Cytology q3y

Stop screening when:

No hx of CIN2+ for
>25 y and adequate*
prior screening x 10 y

* 2 neg HPV ± cytology (<5 y)
OR
3 neg cytology (<3 y)

What if they are post-hysterectomy?

Stop screening if total hysterectomy (cervix removed) for benign
indications

AND

No history of CIN2+ within 25 y

Pathophysiology

Screening

Average risk

High risk

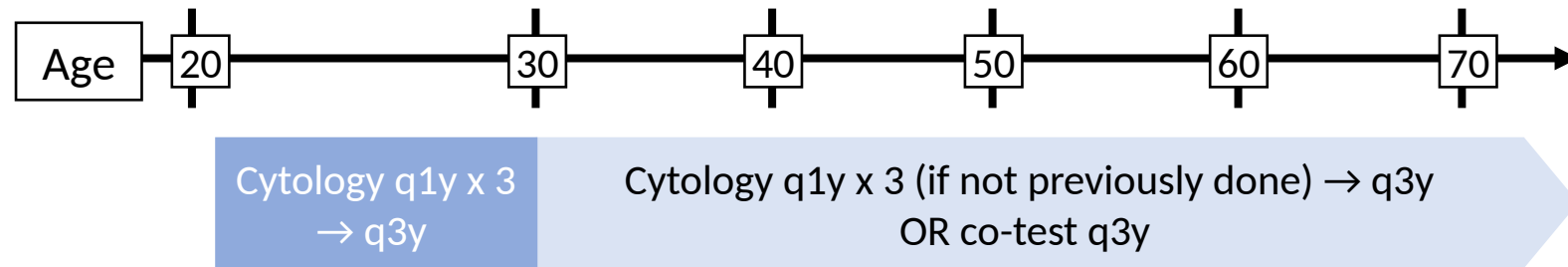
Management

Cases

High-Risk Patients

High risk patients are those with immunocompromise or immunosuppression

- HIV
- Transplant (stem cell, solid organ)
- SLE
- IBD or RA on immunosuppression
- Chronic steroid use

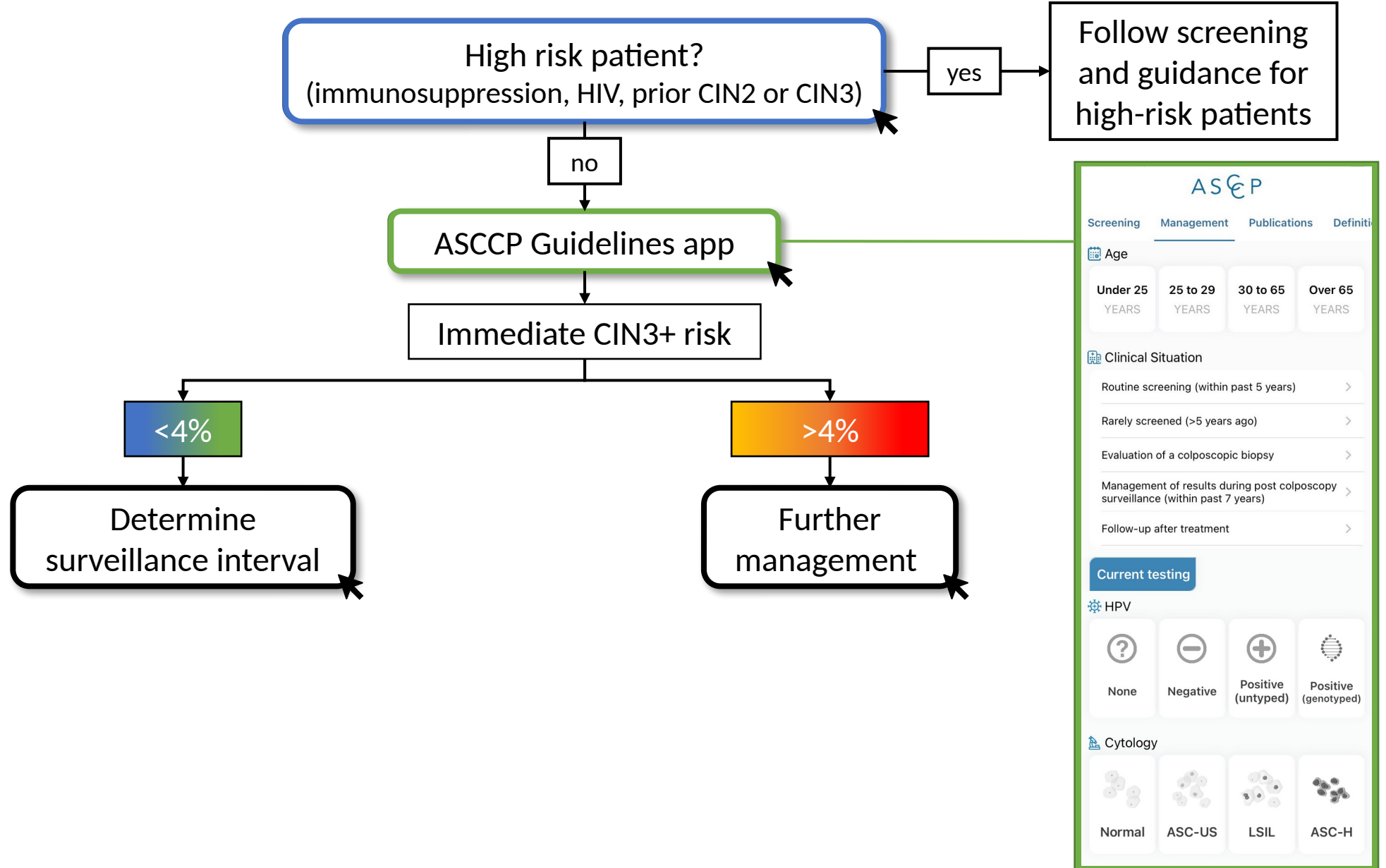
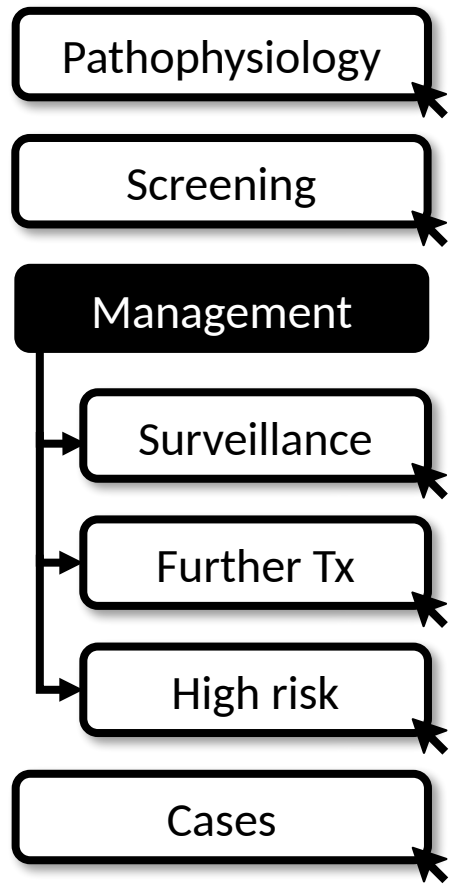


What if they are post-hysterectomy?

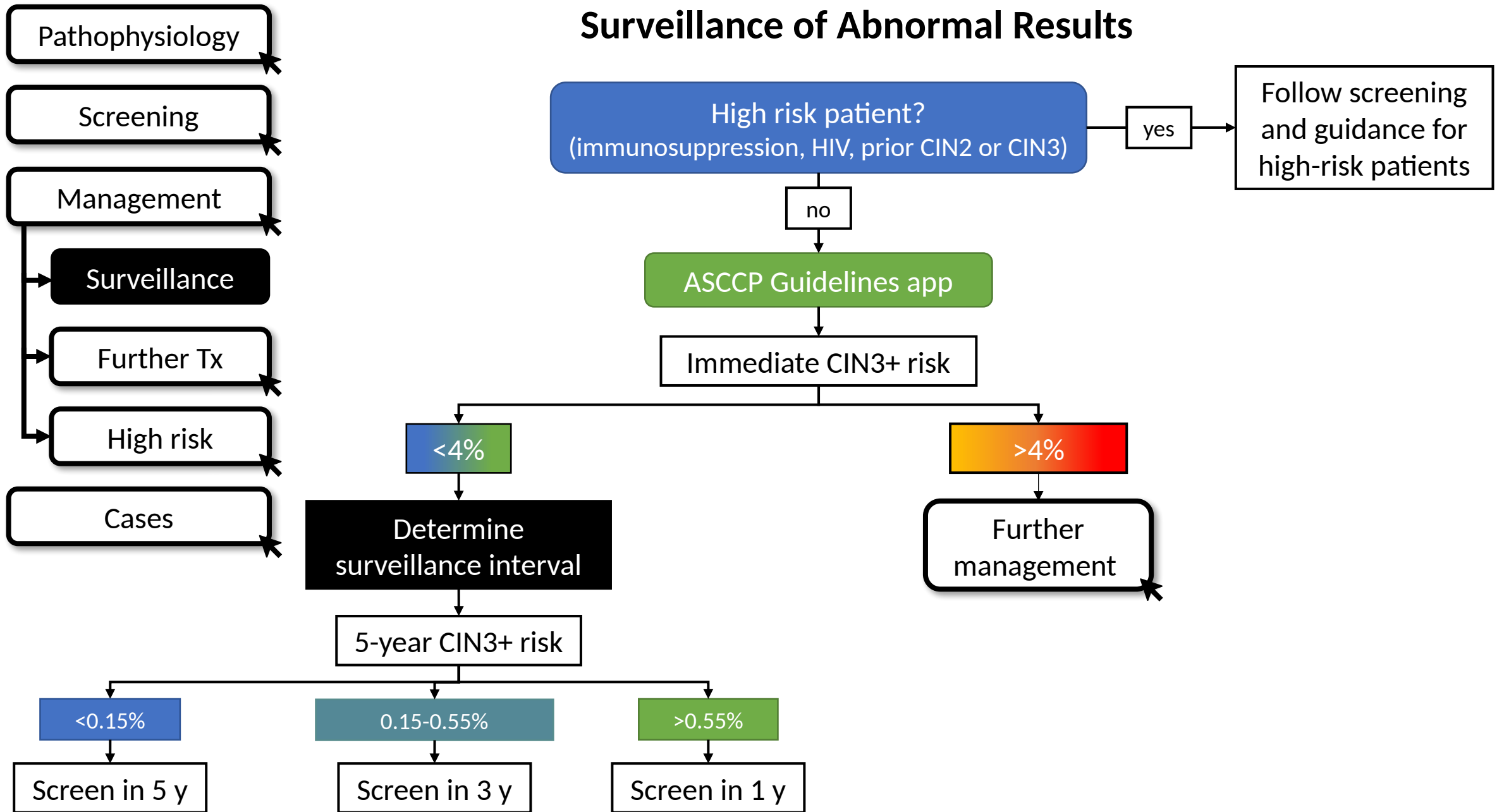
Annual screening (cytology or co-test) if:

- progressive HIV disease
- viral suppression with ART cannot be achieved

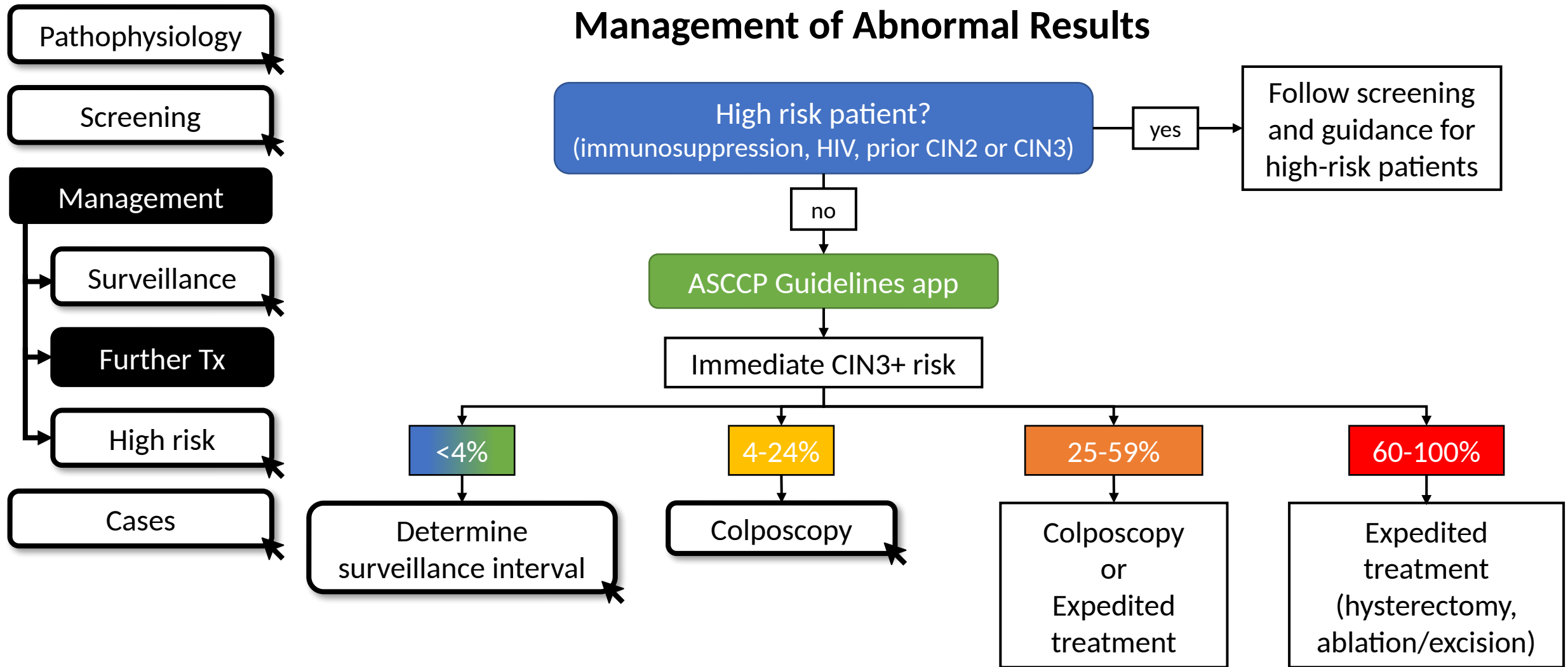
Management of Abnormal Results

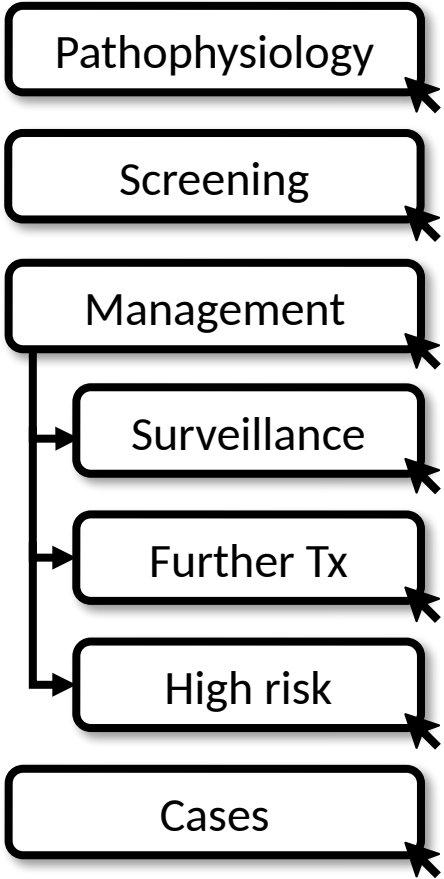


Surveillance of Abnormal Results



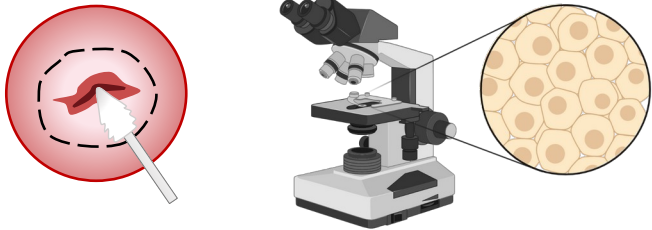
Management of Abnormal Results





Pap smear vs. Colposcopy

Pap smear
(cytology)



Exfoliated **cervical cells** are examined for changes suggestive of acute HPV infection, precancer, or cancer

Colposcopy
(histology)



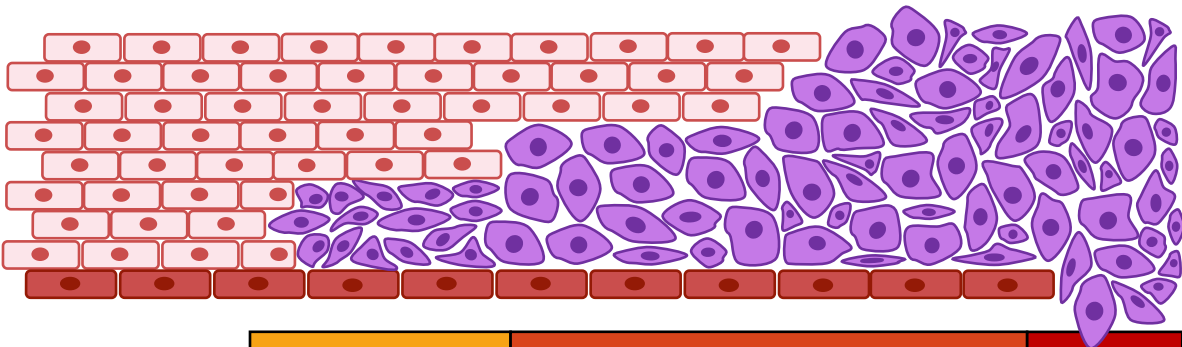
Biopsied tissue architecture is assessed for vertical extension of abnormal cells

Prevalence of Abnormal Cells



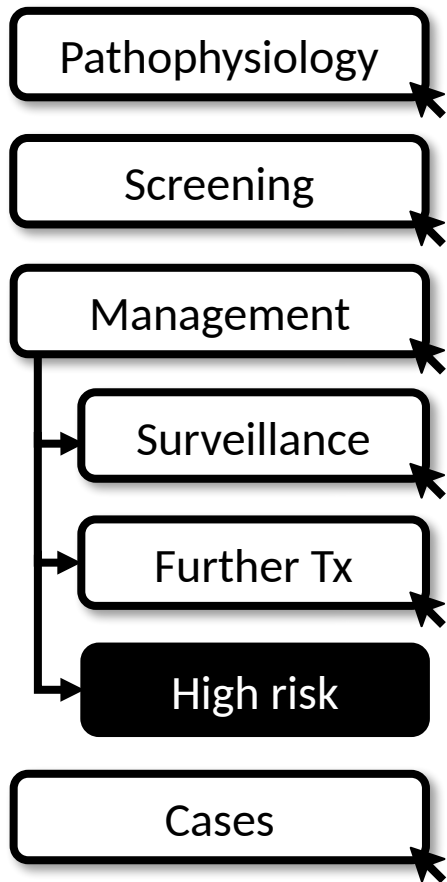
Cytologic findings

NILM	ASC-US	LSIL ASC-H AGC	HSIL
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	Histologic LSIL	Histologic HSIL AIS		Invasive cancer
No CIN	CIN1	CIN2	CIN3	

Histologic findings



Management – High Risk Patients

Immunosuppressed

- HIV
- Solid organ or stem cell transplant
- SLE
- IBD or RA on immunosuppression

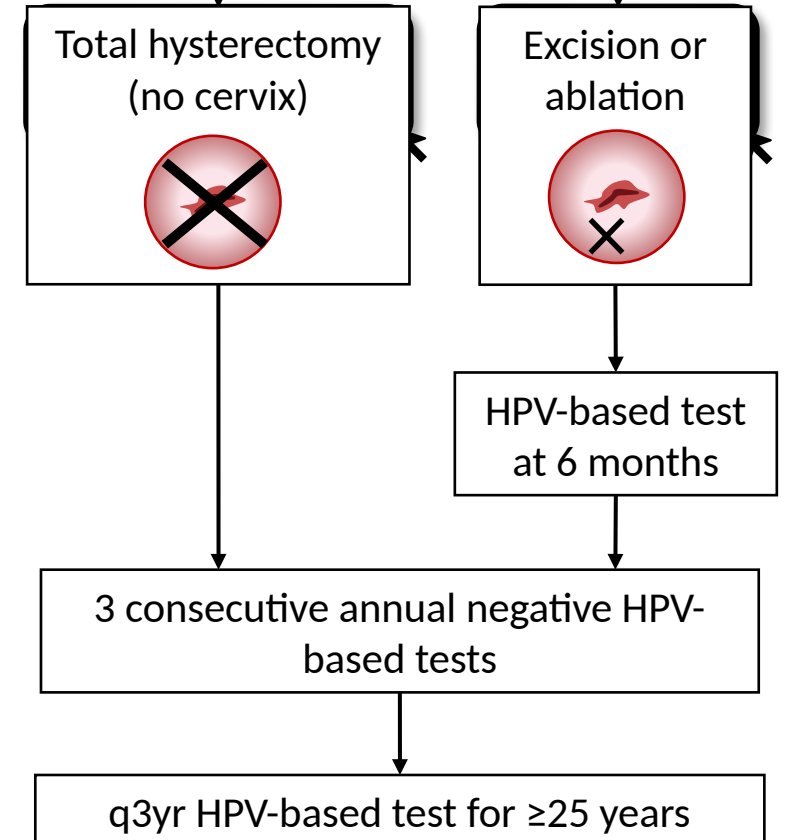
Management

Colposcopy indicated for:

- ASC-US and HPV +
- LSIL | ASC-H | AGC or HSIL regardless of HPV status

History of CIN2 or CIN3

Surveillance



Case 1

Pathophysiology

Screening

Management

Cases

1

2

40 year-old person with a cervix, HTN, depression presents for cervical cancer screening. Their co-test is negative and they have no prior history of abnormal pap smears.

When should they be screened again?

Because the patient had negative HPV-based testing, they should be re-screened in **5 years**.

...Five years later, they return to clinic for repeat screening. Primary HPV testing w/ reflex cytology shows HPV+ for a high risk strain other than 16 or 18, ASC-US.

What is your next step in management or follow-up?

Go to app.asccp.org. Age: 30-65. Clinical situation: Routine screening (within past 5 years). Enter current and prior screening results → **HPV testing at 1 year is recommended.**

...At 1 year follow-up, repeat testing still shows HPV+ other, ASC-US.

What is your next step in management or follow-up?

Go to app.asccp.org. **Colposcopy** is recommended at this point.

...They undergo a colposcopy. Biopsy shows CIN2.

What if your next step in management or follow-up?

Go to app.asccp.org. Age: 30-65. Clinical situation: Evaluation of a colposcopic biopsy. Two options:

- 1) Treatment (excision > ablation) followed by HPV based testing at 6 months is recommended
- 2) For those concerned about impact of treatment on risk of adverse outcomes in future pregnancy, observation with colposcopy + HPV-based screening at 6 and 12 months is acceptable

How would you counsel your patient on the impact of a history of CIN2 on recommended screenings in the future?

- Short-term follow-up: depends on treatment choice.
- Long term follow-up: they are considered high risk and will require screening **at 3-year intervals for a minimum of 25 years (even if >65 years old)**

Pathophysiology

Screening

Management

Cases

1

2

Case 2

37 year-old person with a cervix, SLE on hydroxychloroquine presents for annual wellness exam. They report a normal pap smear 3 y ago but doesn't know if HPV was tested. They completed the HPV vaccine series.

When and how should they be screened for cervical cancer?

Today. Screening can be done with **cytology alone (Pap smear) or HPV co-test at least every 3 years** given history of SLE.

...Pap smear returns with LSIL, HPV negative.

What is your next step in management?

Given their history of immunosuppression, **colposcopy** is indicated despite HPV status.

When can they stop screening?

Given history of SLE on immunosuppression, continue screening every 3 years **throughout the patient's lifetime** (and not, as in the general population, end at 65 years of age), assuming patient is in reasonably good health.

2025 Screening Update — risk-based framework unchanged

- **USPSTF FINALIZED its cervical cancer screening update in 2025 (the deck cites the 2018 version). Average-risk ages 30–65: primary hrHPV every 5 years is now the PREFERRED strategy; cytology q3y remains an option.**
- First-time endorsement of SELF-COLLECTED (patient-collected) hrHPV samples on an FDA-approved platform — non-inferior to clinician-collected (pooled CIN2+ Sn ~0.86, Sp ~0.81); expands access for patients who decline a speculum exam.
- Ages 21–29: cytology alone every 3 years (unchanged). The American Cancer Society (2025) recommends starting at age 25 with primary hrHPV and also endorses self-collection.
- Unchanged: HPV as the cause; ASCCP 2019 risk-based ("equal management for equal risk," 4% immediate-CIN3+ threshold, app.asccp.org) for abnormal results; distinct protocols for HIV/immunosuppressed; stop at 65 if adequate prior screening and no CIN2+ within 25 y.
 - Sources: USPSTF Cervical Cancer Screening, final 2025 (Grade A); ACS 2025 guideline update (CA Cancer J Clin); ASCCP Self-Collection Practice Advisory, Oct 2025. Confirm your lab/platform offers an FDA-approved self-collection assay before adopting.

Take-Home Points

- Persistent infection with oncogenic HPV subtypes (chiefly 16 & 18) causes nearly all cervical cancer; >90% arises at the transformation zone, so sampling it is essential.
- Average-risk screening: cytology alone q3y (21–65), or HPV-based testing (primary hrHPV or co-test) q5y (30–65). USPSTF 2025 + ACS prefer primary hrHPV and now endorse self-collection. Stop at 65 with adequate prior screening and no CIN2+ within 25 y.
- Manage abnormal results by the ASCCP 2019 risk-based app: immediate CIN3+ risk <4% → surveillance interval (1/3/5 y by 5-year risk); 4–24% → colposcopy; 25–59% → colposcopy or expedited treatment; ≥60% → expedited treatment.
- Immunocompromised/immunosuppressed patients (HIV, transplant, SLE, IBD/RA on immunosuppression) are higher-risk: start at diagnosis, screen more often (cytology or co-test, not primary HPV in HIV), continue for life, and have a lower threshold for colposcopy.
- History of CIN2+ requires long-term surveillance: HPV-based testing q3y for ≥25 years, even beyond age 65.

Attribution & Changelog

- Original: "Cervical Cancer Screening," TeachIM.org, published Dec 2022 — Rebecca S. Gold MD, Heather Wainstein MD, Meagan Williams MD; Section Ed. Adelaide McClintock MD; Exec. Ed. Yilin Zhang MD.
 - https://teachim.org/teaching_material/cervical-cancer-screening/ — CC BY-NC 4.0 (attribution required, non-commercial). Optimized June 2026. Deck: Cervical-cancer-1.23.pptx (page embed URL).
- **Correction to this copy (all click-to-reveal animations & BioRender graphics preserved):**
 - Slide 10: high-risk colposcopy criteria "LSIL | ASC-H | AGS" → "AGC" (atypical glandular cells; deck slide 9 uses AGC correctly).
- Appended: this changelog + 2025 Screening Update + Take-Home slides. No original content removed; 12 original slides → 15.
- Companion copied & linked (reviewed, unchanged): cervical-cancer-screening_1.23.pdf (printable teaching guide / learner handout). Package: HTML, OnePager PDF, Anki TSV.
- Educational use only — not medical advice; clinician-review before teaching.